Experiment 31: Customer Segmentation Using Clustering

Aim: To group customers based on their purchasing behavior using K-Means clustering.

Algorithm:

1. Import K-Means.
2. Create customer purchasing data.
3. Apply K-Means clustering.
4. Divide customers into three groups.
5. Display customer segment labels.

Code:

```
from sklearn.cluster import KMeans

X = [[10, 100], [15, 150], [20, 200],
    [50, 500], [55, 550], [60, 600],
    [90, 900], [95, 950], [100, 1000]]

model = KMeans(n_clusters=3, random_state=42, n_init=10)
model.fit(X)

print("Customer Segments:")
print(model.labels_)
```

Input:

Purchase data:

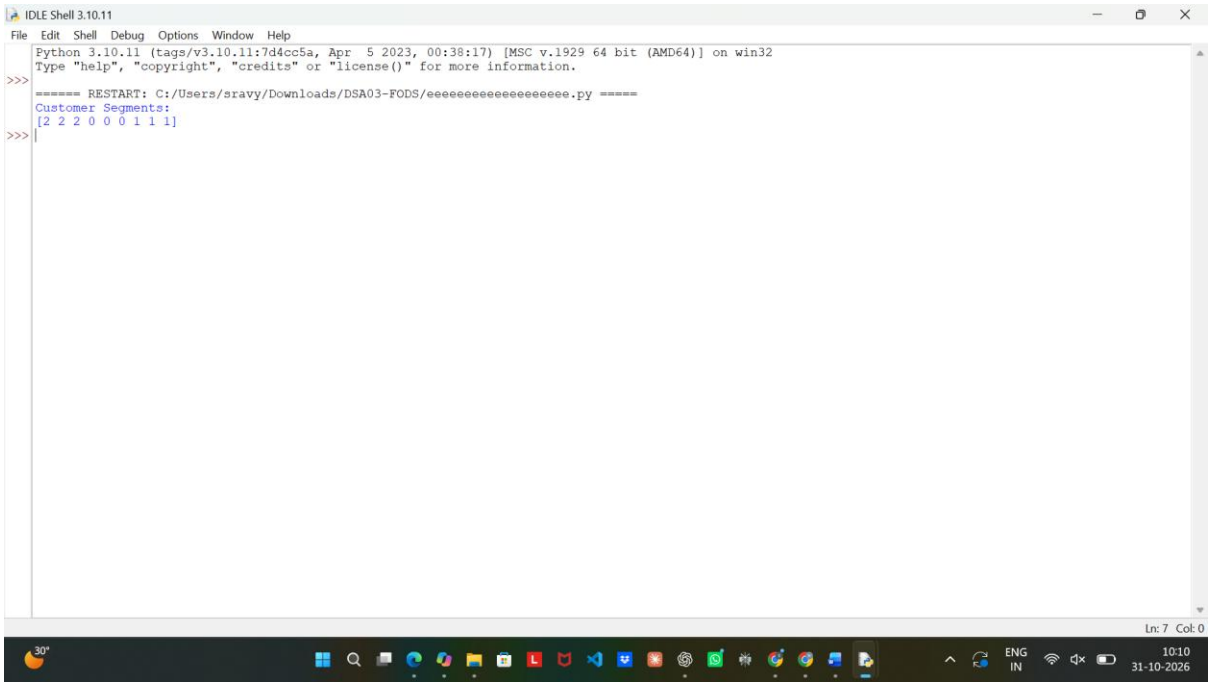
[10, 100], [15, 150], [20, 200]

[50, 500], [55, 550], [60, 600]

[90, 900], [95, 950], [100, 1000]

Output:

```
Customer Segments:
[1 1 1 0 0 0 2 2 2]
```



Result: Customers were successfully grouped into three segments.